

# Day 18 - Lecture 18.15: Adding Annotations to Scatter Plots

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## 1. Overview & Purpose

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In exploratory medical, financial, or performance analysis, each dot in a scatter plot often corresponds to a specific **named entity** (e.g. *Patient A*, *Company X*, *City Y*). Adding **annotations** attaches textual labels directly to coordinates so viewers can instantly identify individual points.

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## 2. Syntax: `plt.annotate()` vs `plt.text()`

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### Method 1: `plt.annotate()` (Recommended for point labeling)

```
plt.annotate(  
    text="Person A",  
    xy=(age, bp),           # Coordinate of the point being annotated  
    xytext=(age + 0.8, bp + 1.2), # Text position (offset to avoid overlapping  
    arrowprops=dict(...))      # Optional pointer arrow  
)
```

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### Preventing Label Clipping with `plt.xlim()`:

When annotating the rightmost point (e.g., Age 65), the text label extends further to the right. Without adjusting axis limits, the text will be cropped!

```
plt.xlim(min(age) - 2, max(age) + 10)
```

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### 3. Code Implementation

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```
import matplotlib.pyplot as plt

people = ["Person A", "Person B", "Person C", "Person D", "Person E",
          "Person F", "Person G", "Person H", "Person I", "Person J"]
age = [22, 25, 30, 35, 40, 45, 50, 55, 60, 65]
blood_pressure = [110, 115, 120, 122, 125, 130, 135, 123, 145, 150]

plt.figure(figsize=(10, 6))
plt.scatter(age, blood_pressure, c=blood_pressure, cmap="OrRd", s=130, edgecolor=
plt.colorbar(label="Blood Pressure (mmHg)")

# Loop through and annotate each person
for i in range(len(age)):
    plt.annotate(
        text=people[i],
        xy=(age[i], blood_pressure[i]),
        xytext=(age[i] + 0.8, blood_pressure[i] + 0.5), # Slight offset to top-right
        fontsize=9,
        fontweight='semibold'
    )

# Add padding to right margin so Person J is not cut off
plt.xlim(min(age) - 3, max(age) + 8)

plt.title("Age vs Blood Pressure with Patient Annotations", fontsize=14, fontweight='bold')
plt.xlabel("Age (Years)", fontsize=12)
plt.ylabel("Blood Pressure (mmHg)", fontsize=12)
plt.grid(True, linestyle=":", alpha=0.6)
plt.show()
```

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### 4. Key Takeaways

- Always apply an offset (  $x + dx$  ,  $y + dy$  ); placing text at the exact point coordinate will superimpose text directly over the marker symbol.
- Use `plt.xlim()` and `plt.ylim()` padding whenever annotations sit near the plot periphery.